



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

EXPERIMENT 1

AIM:

Demonstrate inline, internal, and external JavaScript; use console methods; create a web[page showing user info and a welcome.

Software/Tools Required:

- Visual Studio Code (VS Code)
- Web Browser (Google Chrome/Microsoft Edge)
- HTML
- JavaScript (ES6)

Theory:

JavaScript is a scripting language used to make web pages interactive and dynamic. It can be added to an HTML page in three ways: inline JavaScript (inside HTML elements), internal JavaScript (inside the <script> tag, and external JavaScript (in a separate .js file linked to the HTML page). JavaScript also provides console methods such as console.log(), console.warn(), and console.error() for debugging and displaying messages. In this experiment, JavaScript is used to display user information and a welcome message on a webpage while demonstrating different ways of adding JavaScript to HTML.



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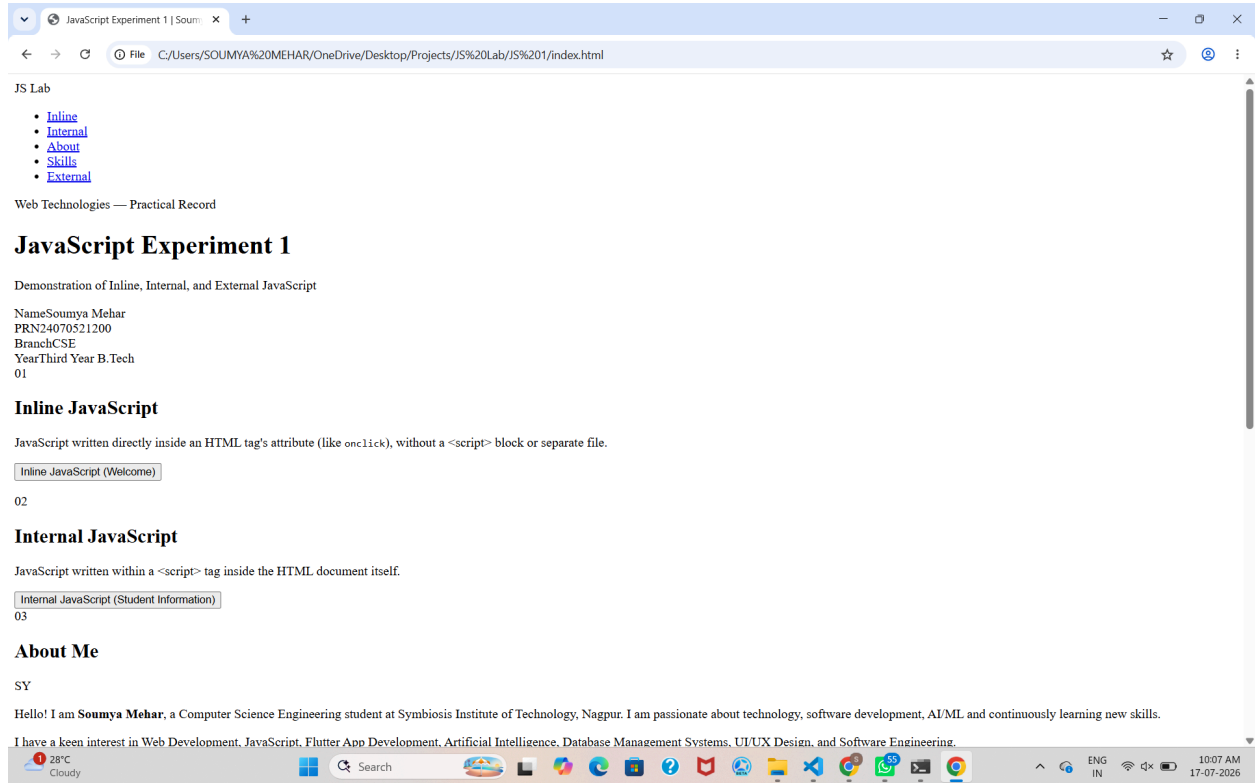
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Output:





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Soumya Mehar

C:/Users/SOUMYA%20MEHAR/OneDrive/Desktop/Projects/JS%20Lab/JS%202/index.html

Soumya Mehar_24070521200

Customer Name:

Add Products

Product Name:

Quantity:

Price per Item (₹):

Added Products

Product Name	Quantity	Unit Price (₹)	Subtotal (₹)	Action
Soumya	1	100.00	100.00	Remove
Mehar	1	500.00	500.00	Remove

Billing Receipt

Customer: 24070521200

Date: 7/17/2026

Product Name	Quantity	Unit Price (₹)	Subtotal (₹)
Soumya	1	100.00	100.00
Mehar	1	500.00	500.00

Grand Total: ₹600.00

GST Amount (18%): ₹108.00

Final Bill Amount: ₹708.00

28°C
Cloudy

Search



ENG
IN

10:15 AM
17-07-2026

Result:

Successfully implemented a billing calculator using JavaScript ES6 features, including var, let, const, template literals, and destructuring, to calculate and display billing details accurately.



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CASE STUDY 1

AIM:

Create a webpage for the college department and link the student information and write a JavaScript program to display “Welcome to SIT Nagpur”.

PROGRAM CODE:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8" />
5   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6   <title>College Department Page</title>
7   <link rel="stylesheet" href="style.css" />
8 </head>
9 <body>
10  <header class="hero">
11    <h1>Department of Computer Science</h1>
12    <p>Symbiosis Institute of Technology, Nagpur</p>
13    <a class="cta" href="student_info.html">View Student Information</a>
14  </header>
15
16  <main class="container">
17    <section class="card">
18      <h2>About the Department</h2>
19      <p>
20        The Computer Science Department is committed to providing quality education,
21        practical learning, and innovation in technology.
22      </p>
23    </section>
24
25    <section class="card">
26      <h2>Key Highlights</h2>
```



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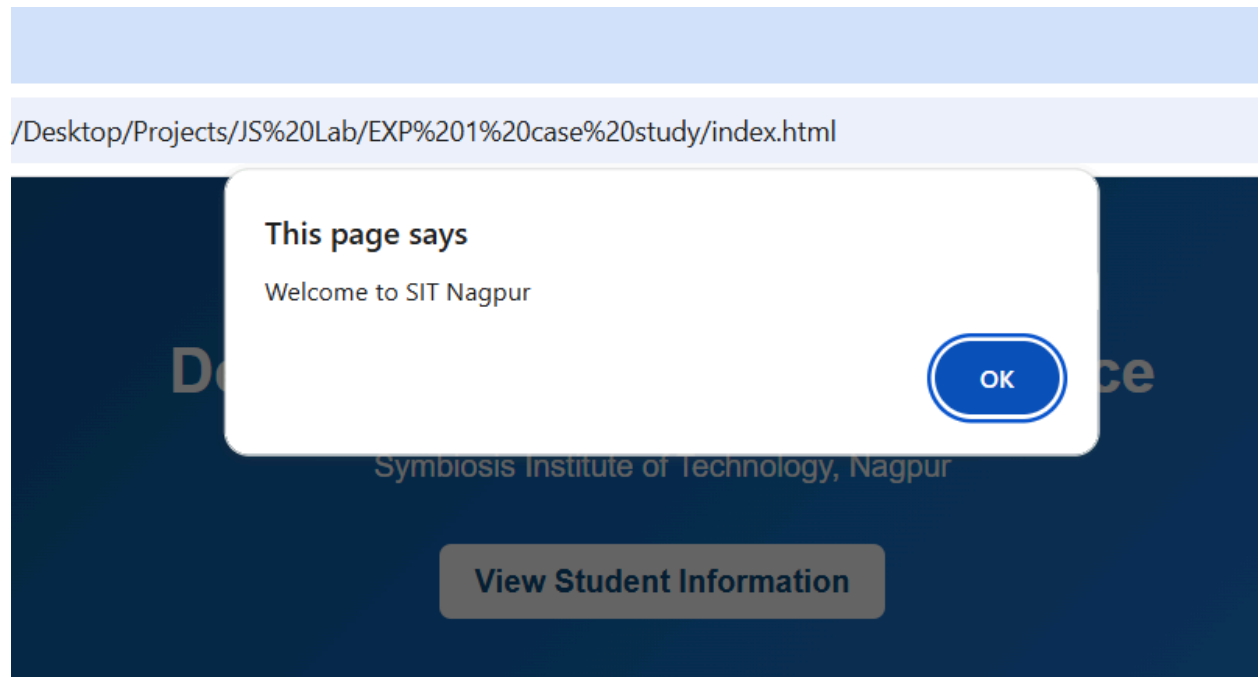
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```
27     <ul>
28         <li>Modern computer labs</li>
29         <li>Experienced faculty members</li>
30         <li>Industry-oriented projects</li>
31         <li>Industry visits</li>
32         <li>Strong placement support</li>
33     </ul>
34 </section>
35 </main>
36
37 <script src="script.js"></script>
38 </body>
39 </html>
```

OUTPUT:





RESULT:

A webpage for the college department was successfully created using HTML, CSS, and JavaScript. Student information was linked and displayed correctly, and the JavaScript program successfully displayed the message **"Welcome to SIT Nagpur"** on the webpage. Thus, the aim of creating a college department webpage and implementing JavaScript functionality was achieved successfully.



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EXPERIMENT 2

AIM:

To use var, let, and const, and demonstrate ES6 features like template literals and destructuring by building a billing calculator.

Software/Tools Required:

- Visual Studio Code (VS Code)
- Web Browser (Google Chrome/Microsoft Edge)
- HTML5
- JavaScript (ES6)

Theory:

JavaScript ES6 (ECMAScript 2015) introduced several modern features that make programming more efficient, readable, and easier to maintain. Among these features are var, let, and const, which are used for variable declaration. The var keyword has function scope and allows variables to be redeclared and updated. The let keyword provides block scope and allows variables to be updated but not redeclared within the same scope. The const keyword is also block-scoped and is used to declare constants whose values cannot be reassigned after initialization.

Template literals are another important ES6 feature that simplify string creation and formatting. They use backticks (```) instead of quotation marks and allow variables and expressions to be embedded directly into strings using `${expression}`. This eliminates the need for string concatenation and improves code readability.

Destructuring is a convenient JavaScript feature that enables the extraction of values from arrays or properties from objects into separate variables. It reduces repetitive code, makes variable assignment more concise, and improves code clarity, especially when working with complex data structures.



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In this experiment, a billing calculator is developed using HTML and JavaScript. The application accepts user inputs such as product details, quantity, and price, and performs calculations to determine the total bill. Variables are declared using var, let, and const wherever appropriate, while template literals are used to display formatted billing information. Destructuring is applied to efficiently access and manage object or array values. This experiment demonstrates the practical implementation of modern JavaScript features in developing a simple, interactive, and user-friendly web application.

Output:

Result:

Successfully implemented a billing calculator using JavaScript ES6 features, including var, let, const, template literals, and destructuring, to calculate and display billing details accurately.



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CASE STUDY 2

AIM:

Develop the webpage for an online shopping website with using html,css and html tags and calculate bill,customer name,mobile no,invoice no,product no,quantity(kg),rate per kg,discount,GST,packing charges,payment mode(cash/card/UPI), Membership(yes/no).

PROGRAM CODE:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <title>Soumya Shopping Cart</title>
7      <style>
8          body {
9              font-family: Arial, sans-serif;
10             background: #f4f8ff;
11             margin: 0;
12             padding: 20px;
13         }
14
15         .container {
16             max-width: 1000px;
17             margin: auto;
18             background: white;
19             padding: 25px;
20             border-radius: 12px;
21             box-shadow: 0 4px 12px rgba(0,0,0,0.08);
22         }
23
24         h1 {
25             text-align: center;
26             color: #2c5aa0;
27         }
```



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```
119 </head>
120 <body>
121   <div class="container">
122     <h1>Soumya Shopping Cart</h1>
123
124     <div class="grid">
125       <div class="field">
126         <label>Customer Name</label>
127         <input type="text" id="customerName">
128       </div>
129
130       <div class="field">
131         <label>Mobile No</label>
132         <input type="text" id="mobileNo">
133       </div>
134
135       <div class="field">
136         <label>Invoice No</label>
137         <input type="text" id="invoiceNo">
```



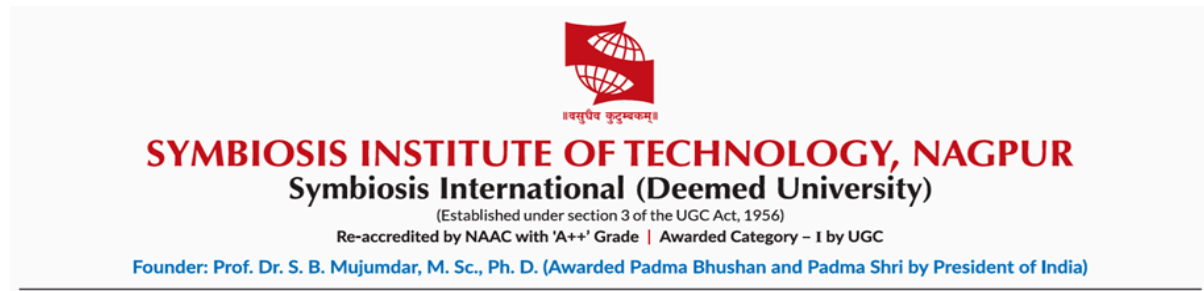
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```
134
135     <div class="field">
136     |   <label>Invoice No</label>
137     |   <input type="text" id="invoiceNo">
138     </div>
139
140     <div class="field">
141     |   <label>Packing Charges</label>
142     |   <input type="number" id="packing" min="0" value="0">
143     </div>
144
145     <div class="field">
146     |   <label>Payment Mode</label>
147     |   <select id="paymentMode">
148     |   |   <option>Cash</option>
149     |   |   <option>Card</option>
150     |   |   <option>UPI</option>
151     |   </select>
152     </div>
153
154     <div class="field">
155     |   <label>Membership</label>
156     |   <select id="membership">
```



OUTPUT:

The screenshot shows a web browser window with the title 'Soumya Shopping Cart'. The URL in the address bar is 'C:/Users/SOUMYA%20MEHAR/OneDrive/Desktop/Projects/JS%20Lab/Case%20study%202/index.html'. The web page has a light blue background. The main content area is a white box with the title 'Soumya Shopping Cart' in blue. It contains several input fields for customer details: 'Customer Name' (Soumya), 'Mobile No' (9588446174), 'Invoice No' (24070521200), 'Packing Charges' (2), 'Payment Mode' (Cash), and 'Membership' (Yes). Below these is a table with columns: 'Product Name', 'Quantity (kg)', 'Rate per kg', 'Discount (%)', and 'GST (%)'. The table has one row for 'Sugar' with values: 5, 10, 20, and 0.18. There is a 'Remove' button next to the row. Below the table are 'Add Item' and 'Calculate Bill' buttons. At the bottom, there is a summary table with the following data: Customer Name: Soumya, Mobile No: 9588446174, Invoice No: 24070521200, Packing Charges: 2.00, Payment Mode: Cash, Membership: Yes, Sugar (5 kg): 40.07, and Total Bill: 42.07. The Windows taskbar is visible at the bottom with various icons and the system clock showing 08:19 PM on 05-08-2026.

RESULT:

The online shopping billing webpage was successfully developed using HTML, CSS, and JavaScript. The program accepted customer and product details, calculated the total bill including discount, GST, and packing charges, and displayed the final invoice correctly. It demonstrated the use of user input, arithmetic calculations, objects, template literals, and object destructuring in JavaScript.



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EXPERIMENT 3

AIM:

To implement JavaScript control structures and form validation by creating a grading system based on user-entered marks.

Software/Tools Required:

- Visual Studio Code (VS Code)
- Web Browser (Google Chrome/Microsoft Edge)
- HTML5
- JavaScript (ES6)

Theory:

JavaScript is a widely used scripting language that enables interactive and dynamic behavior on web pages. One of its key features is the use of control structures, such as if, else if, else, and switch, which allow programs to make decisions based on specified conditions. These control structures help in executing different blocks of code depending on the values provided by the user, making applications more intelligent and responsive.

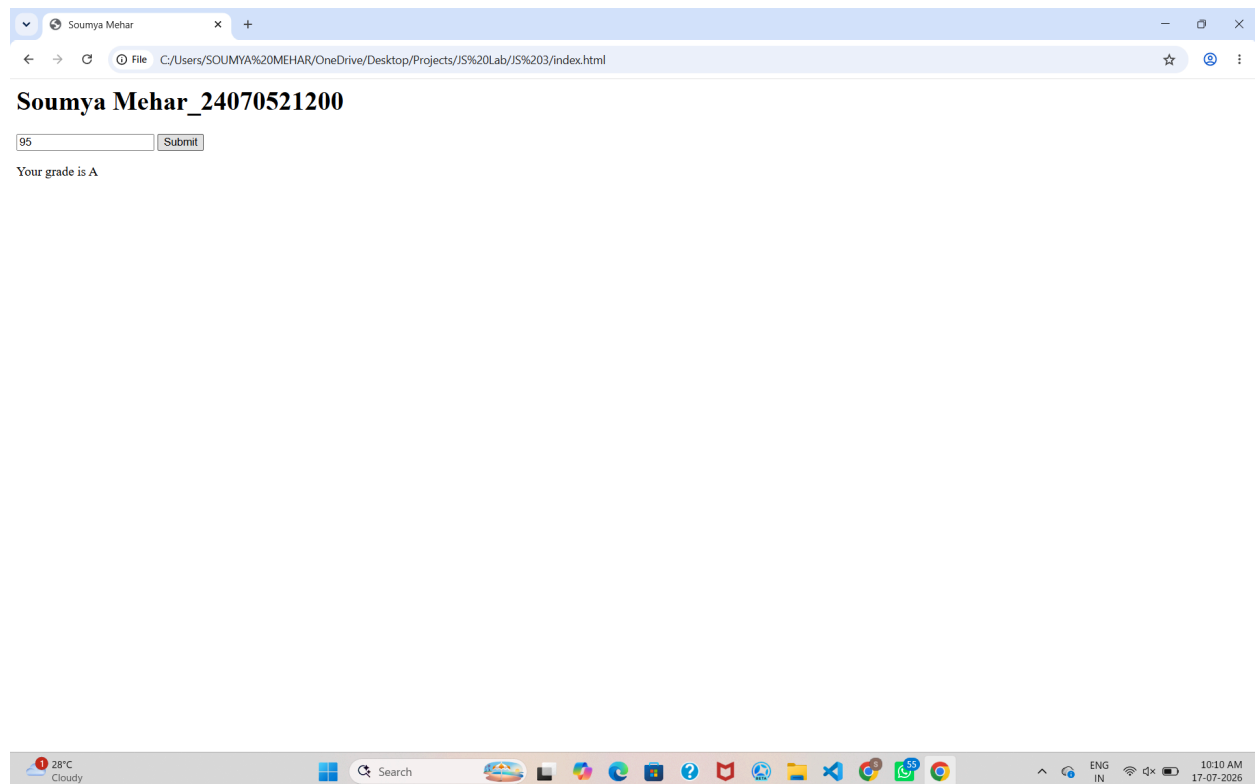
Another important concept in web development is form validation, which ensures that user input is accurate, complete, and within the expected range before it is processed. Validation helps prevent invalid or incomplete data from affecting the application's functionality. It checks conditions such as empty input fields, non-numeric values, and marks entered outside the valid range (0–100). This improves data integrity and enhances the overall user experience.

In this experiment, a grading system is developed using HTML and JavaScript. The user enters marks for different subjects through a form, and JavaScript validates the input before performing calculations. After successful validation, the program computes the total marks, percentage, grade, and pass/fail status. The grade is determined using conditional statements based on predefined percentage ranges, while the pass/fail status is decided according to the minimum passing criteria. This experiment demonstrates the



practical application of JavaScript control structures and form validation in building an interactive and reliable web application.

Output:



Result:

Successfully implemented JavaScript control structures and form validation to create a grading system that validates user input, calculates the total marks, percentage, grade, and pass/fail status accurately.



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CASE STUDY 3

AIM:

Scholarship Eligibility, A college awards scholarships based on marks. Below 50-Not Eligible, 50-69-Bronze, 70-84-Silver, 85-94-Gold, 95-100-Platinum.

PROGRAM CODE:

```
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4    <meta charset="UTF-8" />
5    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6    <title>Soumya Eligibility Tester</title>
7    <style>
8      body {
9        font-family: Arial, sans-serif;
10       margin: 0;
11       padding: 40px 20px;
12       background: #f4f6f8;
13       display: flex;
14       justify-content: center;
15       align-items: center;
16       min-height: 100vh;
17     }
18
19     .card {
20       background: #fff;
21       padding: 24px;
22       border-radius: 12px;
23       box-shadow: 0 4px 12px rgba(0, 0, 0, 0.08);
24       width: 100%;
25       max-width: 420px;
26     }
```



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```
28 h1 {
29   font-size: 24px;
30   margin-bottom: 16px;
31   text-align: center;
32 }
33
34 label {
35   display: block;
36   margin-bottom: 8px;
37   font-weight: bold;
38 }
39
40 input {
41   width: 100%;
42   padding: 10px;
43   margin-bottom: 12px;
44   box-sizing: border-box;
45 }
46
47 button {
48   width: 100%;
49   padding: 10px;
50   border: none;
```




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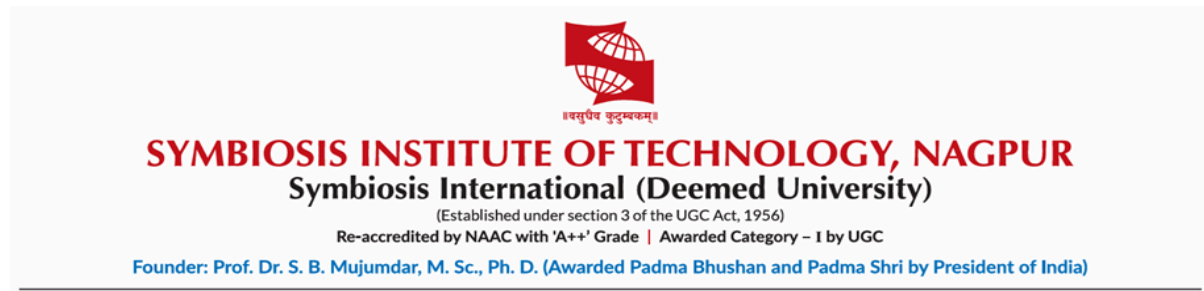
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```
67 <body>
68   <div class="card">
69     <h1>Soumya Eligibility Tester</h1>
70     <label for="marks">Enter marks</label>
71     <input type="number" id="marks" min="0" max="100" placeholder="e.g. 88" />
72     <button onclick="checkScholarship()">Check</button>
73     <div class="result" id="result">Result will appear here</div>
74   </div>
75
76   <script>
77     function checkScholarship() {
78       const marks = Number(document.getElementById('marks').value);
79       let message = 'Please enter a valid mark between 0 and 100.';
80
81       if (marks >= 0 && marks <= 100) {
82         if (marks < 50) {
83           message = 'Not Eligible';
84         } else if (marks <= 69) {
85           message = 'Bronze';
86         } else if (marks <= 84) {
87           message = 'Silver';
88         } else if (marks <= 94) {
89           message = 'Gold';
90         } else {
```



OUTPUT:

The screenshot shows a web browser window with a single tab titled 'Soumya Eligibility Tester'. The address bar shows the file path 'C:/Users/SOUMYA%20MEHAR/OneDrive/Desktop/Projects/JS%20Lab/Case%20Study%203/index.html'. The main content area displays a white box with the title 'Soumya Eligibility Tester'. Inside the box, there is a label 'Enter marks' above a text input field containing the number '40'. Below the input field is a blue button labeled 'Check'. At the bottom of the box is a light blue button labeled 'Not Eligible'. The browser's taskbar at the bottom shows the Windows logo, a search bar, and various application icons. The system tray on the right indicates the temperature is 29°C, the weather is 'Mostly cloudy', and the time is 08:30 PM on 05-08-2026.

RESULT:

The Scholarship Eligibility was successfully implemented using JavaScript. The program validated the user input, assigned eligibility marks using if-else control structures, and displayed the correct Eligibility awards based on the entered marks. It effectively demonstrated the use of control structures and form validation in web development.



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EXPERIMENT 4

AIM:

Use function types, scope and closures; apply try-catch; build a palindrome checker

Software/Tools Required:

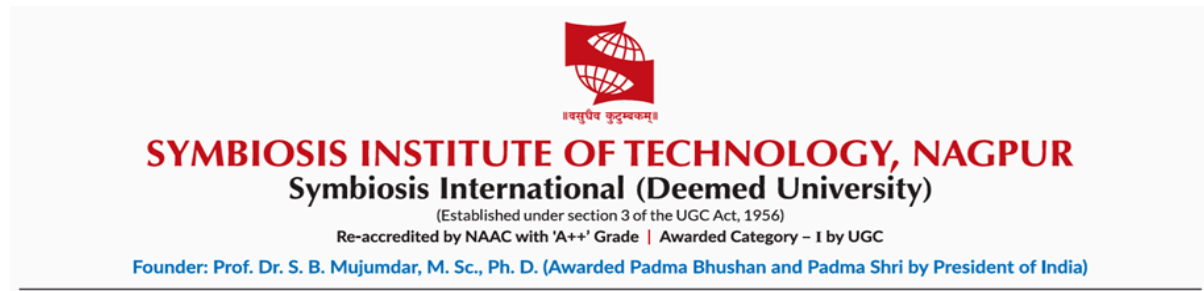
- Visual Studio Code (VS Code)
- Web Browser (Google Chrome/Microsoft Edge)
- HTML
- JavaScript (ES6)

Theory:

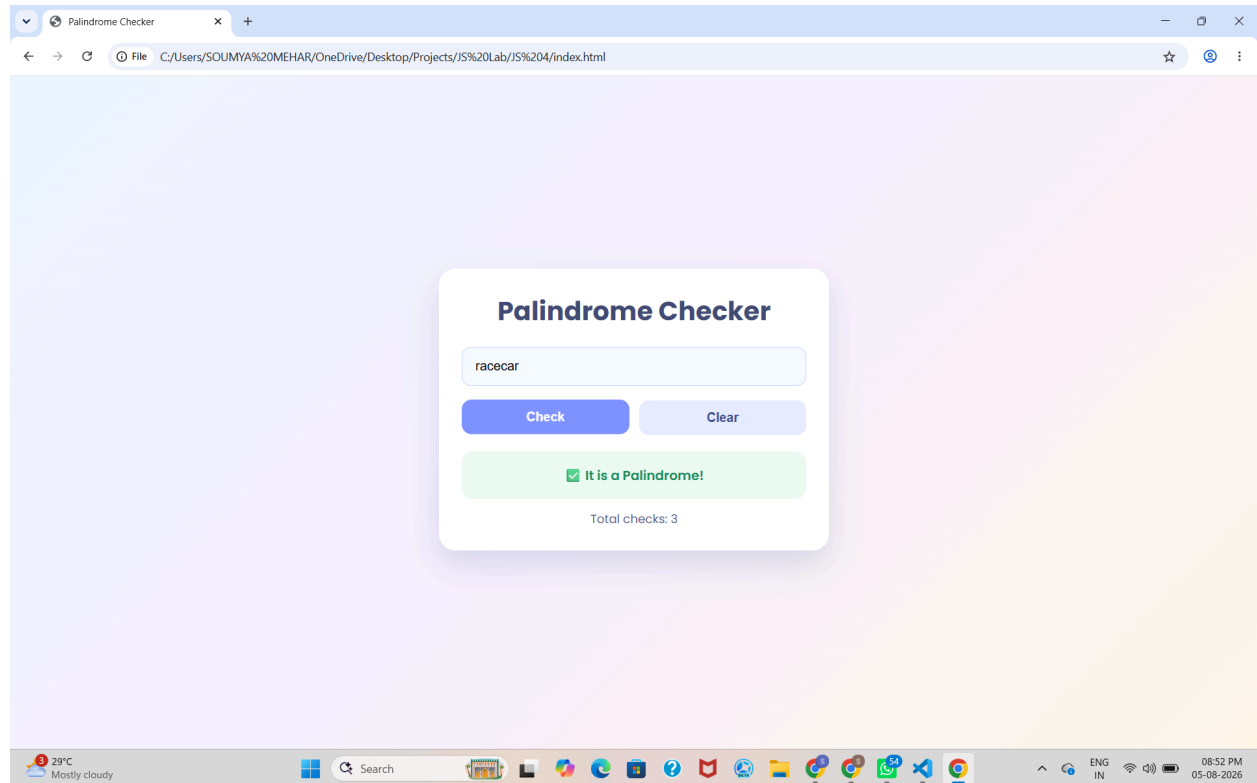
A palindrome is a word, number, or sentence that reads the same forward and backward, such as MADAM, LEVEL, and 121. A palindrome checker is a program that checks whether the given input is a palindrome or not by comparing the original string with its reverse.

In JavaScript, different types of functions, such as function declarations, function expressions, and arrow functions, can be used to implement the palindrome checker. Variable scope determines where variables can be accessed in the program, while closures allow an inner function to access variables from its outer function.

The try-catch statement is used for exception handling and helps manage errors, such as invalid or empty input, without stopping the execution of the program. Thus, a palindrome checker demonstrates the concepts of functions, scope, closures, and error handling in JavaScript.



Output:



Result:

Successfully implemented a Palindrome detector using JavaScript ES6 features, including var, let, const, template literals, and destructuring, to find and display whether the given text or number is a palindrome or not.



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CASE STUDY 4

AIM:

The ATM system verifies if the pin is palindrome. If the palindrome displays a security message. To implement this create a function to reverse the entered pin number and another function to check if it is palindrome. use function declaration, function expression, arrow function and demonstrate scope and closure concepts.

PROGRAM CODE:

```
1  <!DOCTYPE html>
2  <html lang="en">
3    <head>
4      <meta charset="UTF-8" />
5      <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6      <title>ATM Palindrome PIN Checker</title>
7    </head>
8    <body>
9      <div>
10        <div>
11          <div>
12            <div>
13              <div>
14                <div>
15                  <div>
16                    <div>
17                      <div>
18                        <div>
19                          <div>
20                            <div>
21                              <div>
22                                <div>
23                                  <div>
24                                    <div>
25                                      <div>
26                                        <div>
27                                          <div>
```



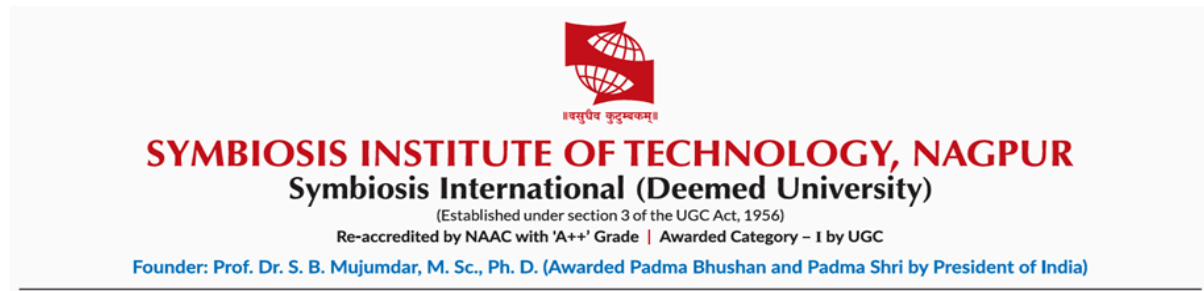
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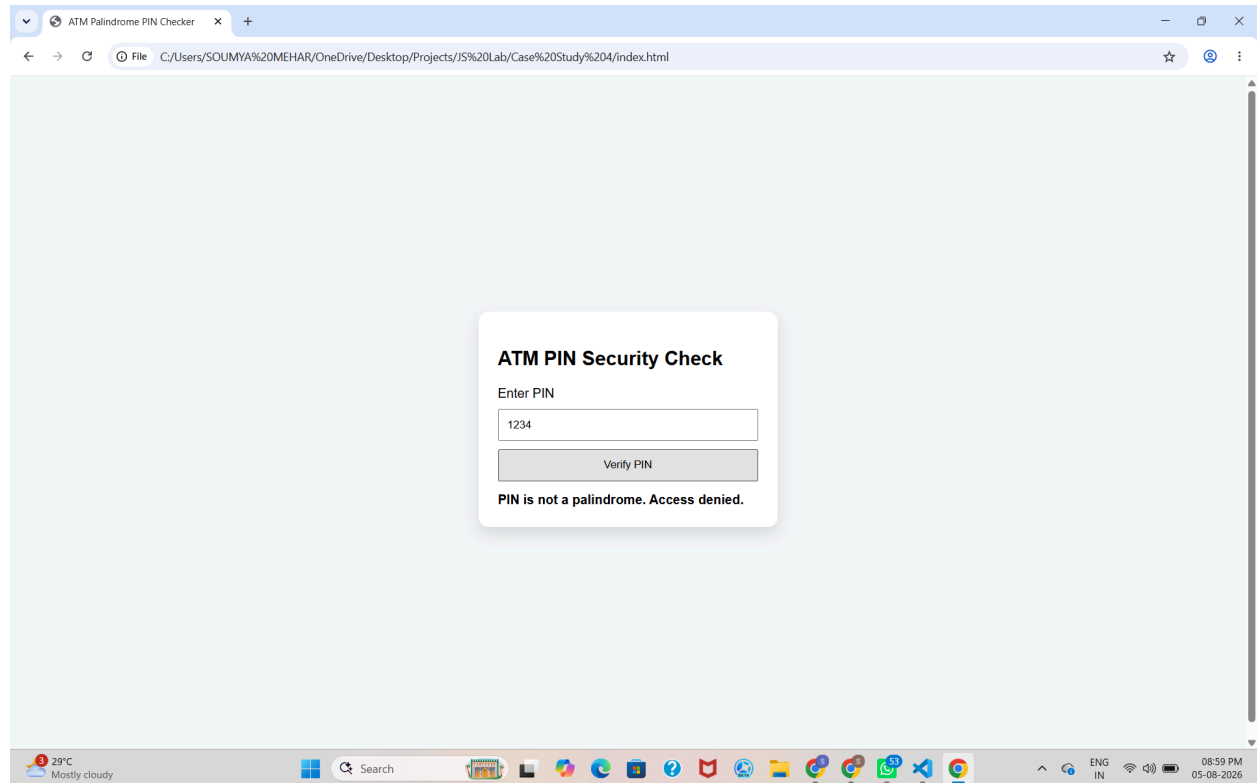
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```
28     }
29     #result {
30     |   margin-top: 14px;
31     |   font-weight: bold;
32     |   }
33   </style>
34 </head>
35 <body>
36   <div class="card">
37     <h2>ATM PIN Security Check</h2>
38     <label for="pinInput">Enter PIN</label>
39     <input id="pinInput" type="text" placeholder="e.g. 121" />
40     <button id="checkBtn">Verify PIN</button>
41     <div id="result">Result will appear here</div>
42   </div>
43
44   <script src="script.js"></script>
45 </body>
46 </html>
47
```



OUTPUT:



RESULT:

Successfully implemented a ATM pin Palindrome detector using JavaScript ES6 features, including var, let, const, template literals, and destructuring, to find and display whether the given ATM pin is a palindrome or not.